

Dear client,

Thank you for purchasing this market data delivery. The data is based on TradingView market data and is delivered fully processed and ready to use.

Order details

Assets:

NBTB LMT NOC HON LHX PKG GWW CELC COST ODFL MSCI URNU QNCCF

Indicators:

RSI, MACD, EMAs, Bollinger Bands

Timeframes:

1D

Delivered file formats

All file formats include the same data. Only the data format changes. You can choose any file format with confidence, as they all deliver identical data.

The Excel files use a more human-friendly header naming.

The JSON and CSV files use the same 1:1 field naming making it more suitable for code-oriented and programmatic use.

XLSX (Excel)

The Excel file provides a visual and exploratory representation of the dataset, organized into multiple worksheets by data category.

Formatting features such as freeze panes, cell borders, header styling, and value formatting are applied by default to improve readability and navigation.

Workbook structure

Each worksheet represents a specific data group derived from the same underlying candle structure.

The workbook includes the following sheets:

- OHLCV
- Price and Volatility
- Volume
- RSI
- MACD
- EMAs
- Bollinger Bands

Each sheet focuses on a single data category to keep the file readable and avoid overcrowded tables.

Common base columns

All sheets share the same base identification columns:

- Symbol
- Timeframe
- Timestamp
- Formatted Date

These columns provide full context for every row and allow sheets to be used independently.

Each row represents one fully closed candle.

Sheet independence

Each worksheet can be analyzed on its own without requiring cross-references to other sheets.

This design allows: - Filtering, sorting, and charting within a single sheet. - Easy comparison across time without losing asset or timeframe context. - Selective export or copy of individual sheets if needed.

Data consistency

- All sheets are aligned by candle timestamp.
- No data is duplicated across sheets beyond the shared base columns.
- The values shown in Excel match exactly the data provided in JSON and CSV formats.

For detailed explanations of the metrics included in each sheet, refer to the **Understanding Metrics** section.

JSON structure

The JSON file contains two top-level keys: `meta` and `data` .

Top-level keys

- `meta` : Global metadata for the dataset.
 - `data` : Time-series container holding candle entries.
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`meta`

Flat object containing dataset-wide attributes.

Fields: - `symbol` - `timeframe` - `candles`

All values in `meta` apply uniformly to every candle entry in `data` .

`data`

`data` is an object indexed by Unix timestamps (seconds).

- Each key is a Unix timestamp.
- Each value represents one fully closed candle.
- Each key maps 1:1 to the internal `timestamp` field inside the candle object.

Candle structure

Each candle groups related data into dedicated nested objects to maintain a clean and predictable structure.

Each candle entry contains:

- `timestamp`
- `formatted_date`
- `ohlcv`
- `price_and_volatility`
- `volume`
- `indicators`

The `ohlcv`, `price_and_volatility`, and `volume` objects contain their related fields grouped by category.

For detailed explanations of the metrics included in these objects, refer to the **Understanding Metrics** section.

`indicators`

Container object for all technical indicator data.

- Each key inside `indicators` represents one indicator.
- Each indicator key maps to an object containing only that indicator's fields.
- Only indicators requested in the order are included.

No indicator fields exist outside the `indicators` object.

Structural guarantees

- One candle equals one timestamp entry.
- All candles share the same schema.
- All derived data is strictly grouped under its corresponding container.
- The structure is stable and consistent across all entries.
- Key names follow a consistent snake_case naming convention.

CSV structure

The CSV file provides the same data as the XLSX and JSON formats, flattened into a single tabular structure.

Each row represents one fully closed candle.

All data is stored at a single column level, with no nested structures.

Row structure

- One row equals one candle.
 - There is exactly one row per candle.
 - Rows are ordered chronologically.
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Column structure

The CSV uses a flat and explicit column layout.

Base identification columns

These columns uniquely identify each candle and provide context:

- `symbol`
 - `timeframe`
 - `timestamp`
 - `formatted_date`
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Data columns

All remaining columns represent candle data flattened from the JSON structure.

This includes: - OHLCV fields - Price, volatility, and volume-derived fields - Indicator fields

All columns exist at the same level.

No fields are nested or grouped in the CSV format.

For detailed explanations of individual metrics and indicator fields, refer to the **Understanding Metrics** section.

Structural characteristics

- One row equals one candle.
 - The CSV is fully denormalized into a single table.
 - Column names follow a consistent snake_case naming convention.
 - The file is suitable for direct ingestion into scripts, databases, and analytical tools.
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Naming and metric consistency

As stated previously, all delivered file formats contain exactly the same underlying data and metrics.

The only difference between formats is how that data is presented.

The JSON and CSV files use identical field and key names, following a consistent snake_case naming convention.

The Excel (XLSX) file presents the same metrics using a more human-readable naming style and a visual layout optimized for manual analysis. Despite these visual differences, the underlying metrics are the same across all formats.

No metric exists in one format that does not exist in the others.

No metric is calculated differently between formats.

With this clarification in mind, the following section provides a detailed explanation of each metric included in the delivery.

Understanding metrics

OHLCV

Standard OHLCV data for each candle. In addition to volume expressed in units (shares), this dataset also includes `volume_usd`, representing traded volume expressed in USD.

Price and Volatility

Important note on timeframe-dependent naming

Price change, volatility, and range metrics are always calculated over a fixed number of candles.

However, the naming of these metrics adapts depending on the selected timeframe.

- For **daily timeframes** (e.g. `1D`), metrics are expressed using calendar-based labels such as `24h`, `7d`, `14d`, and `30d`.
- For **intraday timeframes** (e.g. `1H`, `4H`), the same metrics are expressed using the number of candles instead of calendar units.

Despite the different naming, the underlying logic remains the same:

all values represent changes, volatility, or ranges computed over a specific number of consecutive candles.

Example

If the selected timeframe is `1D` : - `price_change_7d_pct` represents the price change over the last 7 daily candles.

If the selected timeframe is `1H` : - `price_change_last_7_candles_pct` represents the price change over the last 7 hourly candles.

The same rule applies to volatility and range metrics, which are always calculated over a defined number of candles, regardless of timeframe.

Price change metrics

Price change metrics represent percentage price deltas computed over a fixed number of consecutive candles.

For each candle, the price change is calculated as the percentage difference between the current close price and the close price `N` candles ago.

The same calculation logic is used for all window sizes (7, 14, and 30); only the number of candles considered changes.

Values are expressed as percentages.

Volatility metrics

Volatility metrics represent the dispersion of price returns over a fixed number of consecutive candles.

Volatility is computed as the standard deviation of percentage price returns over the selected window size.

As with price change metrics, the same calculation logic applies to all window sizes (7, 14, and 30); only the number of candles included in the window changes.

Values are expressed as percentage points.

Range metrics

Range metrics measure the price movement within a single candle.

The range is calculated as the percentage difference between the high and low of the candle, normalized by the opening price.

For daily timeframes, this metric is exposed as `intraday_range_pct`.

For intraday timeframes, the same metric is exposed as `candle_range_pct`.

Despite the different naming, both fields represent the same underlying calculation.

Volume

This section contains volume-derived metrics. Raw volume values are available in the OHLCV data.

Important note on timeframe-dependent naming

Volume change metrics follow the same timeframe-dependent naming logic described in the *Price and Volatility* section.

- For **daily timeframes** (e.g. `1D`), volume changes are expressed using calendar-based labels such as `24h`, `7d`, `14d`, and `30d`.
- For **intraday timeframes** (e.g. `1H`, `4H`), the same metrics are expressed using the number of candles instead of calendar units.

Despite the different naming, all volume change metrics are computed over a fixed number of consecutive candles.

Volume change metrics

Volume change metrics represent percentage deltas of traded volume over a fixed number of candles.

For each candle, the volume change is calculated as the percentage difference between the current candle's volume and the volume `N` candles ago.

The same calculation logic is applied across all window sizes (7, 14, and 30); only the number of candles included in the comparison changes.

Values are expressed as percentages.

Volume vs average metrics

Volume vs average metrics compare the current candle's volume against the average volume of a fixed window.

The average is computed over the last `N` candles, and the resulting value represents the percentage difference between the current volume and that rolling average.

For daily timeframes, this metric is exposed as: - `volume_vs_30d_avg_pct`

For intraday timeframes, the same metric is exposed as: -

`volume_vs_30_candles_avg_pct`

Despite the different naming, both fields represent the same underlying calculation.

Volume units

All volume-based metrics are derived from traded volume data.

- Raw volume values are available as `volume_units` (shares) and `volume_usd`.
 - All volume change and comparison metrics are computed using the same underlying volume data.
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RSI

The RSI indicator exposes a single value per candle.

- `value` represents the RSI value calculated for that candle.
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MACD

The MACD indicator exposes three values per candle.

- `macd_line` : corresponds to the main MACD line (commonly displayed as the blue line in charting interfaces).
- `signal_line` : corresponds to the signal line (commonly displayed as the orange line).
- `histogram` : corresponds to the histogram bars, representing the difference between the MACD line and the signal line.

Field naming follows the standard MACD representation commonly used in charting platforms such as TradingView.

EMAs

The EMAs indicator exposes exponential moving average values and derived gap metrics for each candle.

EMA values

EMA values are provided for the following periods:

- `ema_20`
- `ema_50`
- `ema_100`
- `ema_200`

Each field represents the EMA value calculated for that candle using the corresponding period.

Gap vs EMA metrics

For each EMA, a corresponding gap metric is provided:

- `gap_ema_20_pct`
- `gap_ema_50_pct`
- `gap_ema_100_pct`
- `gap_ema_200_pct`

Gap metrics represent the percentage difference between the candle close price and the corresponding EMA value.

The gap is calculated as: $(\text{close} - \text{EMA}) / \text{EMA} * 100$

- A positive value indicates that the close price is above the EMA.
- A negative value indicates that the close price is below the EMA.

All gap values are expressed as percentages and are calculated independently for each candle.

Bollinger Bands

The Bollinger Bands indicator exposes the main band values and derived metrics for each candle.

Bollinger Bands values include:

- `bb_mid` : middle band value.
- `bb_upper` : upper band value.

- `bb_lower` : lower band value.

Each field represents the corresponding Bollinger Band value calculated for that candle.

In addition to the band values, the following derived metrics are provided.

Band width

- `band_width_pct`

Band width represents the relative distance between the upper and lower bands, expressed as a percentage.

The band width is calculated as: $(bb_upper - bb_lower) / bb_mid * 100$

This value provides a normalized measure of band expansion or contraction.

Gap vs Bollinger Bands metrics

For each band, a corresponding gap metric is provided:

- `gap_bb_mid_pct`
- `gap_bb_upper_pct`
- `gap_bb_lower_pct`

Gap metrics represent the percentage difference between the candle close price and the corresponding Bollinger Band value.

The gap is calculated as: $(close - band_value) / band_value * 100$

- A positive value indicates that the close price is above the band.
- A negative value indicates that the close price is below the band.

All gap values are expressed as percentages and are calculated independently for each candle.

Scope and intended use

This product provides structured, processed market data only.

It is not financial advice, trading advice, investment research, or a trading system. No signals, recommendations, strategies, or automated decision-making logic are included.

The delivered files contain historical market data and derived metrics intended to be used as inputs for: - Market analysis - Research - Backtesting - Data science workflows - Custom tools or models built by the client

All interpretation, analysis, and decision-making based on this data are the sole responsibility of the user.

Further customization

If you need additional indicators, different timeframes, other assets, or custom data outputs, feel free to reach out.

Custom requests can be discussed to ensure the delivery matches your specific use case.
